Energy- and Carbon-Aware LLM Serving

GreenCache and PRISM

Power-, Energy-, and Carbon-Aware GPU Serving Optimization

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Cache Your Prompt When It's Green — Carbon-Aware Caching for Large Language Model Serving

Carbon-Aware KV Cache Management for LLM Serving

Motivation SSD-based prefix/KV caching avoids redundant prefill and operational carbon from electricity, but adds embodied carbon from SSD manufacturing.

Key Insight Carbon-efficient cache capacity depends on workload intensity and grid carbon intensity.

Method An ILP allocates SSD capacity dynamically. Least Carbon Savings (LCS) selects cache entries for replacement.

Experiments Llama-3 70B (INT8), 4× L40 GPUs, LMCache, up to 16 TB SSD. Workloads: ShareGPT and TriviaQA.

Results In France, 15.1% average carbon reduction vs. Full Cache, peaking at 25.3% in multi-turn serving (70B). Reported latency SLO attainment: $> 90\%$.

Conclusion Carbon-efficient caching balances avoided GPU computation against storage-related carbon costs.

Source: Tian et al., *Cache Your Prompt When It's Green — Carbon-Aware Caching for Large Language Model Serving.* SIGMETRICS 2026 / POMACS 10(1), Article 5. DOI: 10.1145/3788087 Paper PDF Project

Beyond Energy: When Sustainability Dimensions Reshape LLM Serving Decisions

Beyond Energy: Sustainability-Aware LLM Serving

Motivation Across deployment choices, minimizing energy need not minimize carbon, water, or biodiversity impacts.

Key Insight For fixed workload, location and time, operational-only accounting preserves the energy-based computing-configuration ranking in all four dimensions.

Method PRISM evaluates four environmental dimensions and routes across regions to minimize the largest relative penalty (regret) vs. each dimension's optimum.

Experiments Compare GPU/model configurations and regional deployment decisions using environmental-intensity data.

Results 50.2% median reduction in worst-case regret vs. the offline WaterWise adaptation in the 12-region evaluation (20 capacity seeds).

Conclusion Energy guides fixed-site operational optimization. Across regions/time, or with embodied impacts, rankings can differ.

Source: Shi et al., *Beyond Energy: When Sustainability Dimensions Reshape LLM Serving Decisions.* 2026. arXiv:2609.35569 Paper PDF